

BULMAN, AustraliaWMO: **941430**Lat: **13.6714S**Lon: **134.3414E**Elev: **344**StdP: **14.51**Time Zone: **9.50 (AUC)**Period: **04-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	50.1	53.3	34.0	29.1	80.6	37.4	33.3	80.1	14.2	81.9	13.7	82.1	0.6	200	0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	23.6	102.2	71.9	100.6	72.1	99.1	72.1	80.0	90.2	79.2	89.3	78.7	88.6	7.4	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.5	144.9	82.8	76.9	142.0	82.3	76.3	139.2	82.1	43.9	90.5	43.1	89.6	42.5	88.9	90.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.6	12.1	11.1	45.4	104.8	3.9	1.1	42.6	105.6	40.4	106.3	38.2	106.9	35.4	107.7		
(5)			43.7	82.6	4.1	3.3	40.7	85.0	38.3	86.9	36.0	88.7	33.1	91.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.6	84.7	84.3	83.3	80.6	77.1	72.6	72.2	74.1	79.8	84.8	87.5	87.0		
	DBStd	6.40	2.45	2.60	2.44	2.95	4.20	5.26	4.59	4.29	4.06	3.43	2.71	2.99		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	12	0	0	0	0	0	6	5	1	0	0	0	0		
	CDD50	11184	1075	960	1033	918	841	677	689	746	895	1079	1125	1146		
	CDD65	5721	610	540	568	468	376	233	229	282	445	614	675	681		
	CDH74	69656	6999	6076	6121	4987	3826	2276	2543	3686	6121	8761	9397	8863		
	CDH80	35941	3294	2892	2901	2383	1628	792	943	1719	3498	5325	5639	4927		
(14) Wind	WSAvg	4.7	4.4	4.0	3.6	4.0	4.6	5.1	5.2	5.2	5.2	5.3	5.1	4.5		
(15) Precipitation	PrecAvg	47.5	10.5	10.7	8.3	2.6	0.6	0.3	0.2	0.2	0.4	1.2	3.8	8.7		
	PrecMax	72.4	17.2	19.8	18.9	11.9	2.1	0.9	0.9	1.0	3.5	5.7	8.4	13.8		
	PrecMin	31.9	4.3	4.3	1.6	0.1	0.1	0.1	0.0	0.0	0.0	0.0	1.3	3.5		
	PrecStd	10.9	3.8	4.1	4.6	2.6	0.5	0.2	0.2	0.2	0.6	1.3	1.5	3.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	100.2	99.5	98.9	96.3	94.0	91.7	91.6	95.2	99.6	102.9	104.4	104.4		
		MCWB	75.7	75.6	75.0	73.0	72.6	69.7	69.0	68.2	68.8	70.7	72.4	71.7		
	2%	DB	97.5	97.2	96.2	94.5	91.8	89.4	89.5	92.6	97.7	101.0	102.3	101.9		
		MCWB	76.5	76.0	75.3	72.9	71.7	69.4	67.8	67.3	68.1	70.4	71.8	73.2		
	5%	DB	95.2	95.3	94.2	93.0	90.0	87.1	87.6	90.7	96.2	99.6	100.6	99.7		
		MCWB	77.5	76.8	75.6	73.3	71.2	68.3	66.7	66.2	67.6	70.2	71.8	74.2		
	10%	DB	92.8	93.3	92.3	91.0	88.2	84.9	85.6	88.8	94.5	98.0	99.0	97.4		
		MCWB	77.6	77.1	76.0	73.1	70.7	67.4	65.7	65.5	67.1	69.9	72.0	74.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.9	81.0	80.7	78.5	77.7	73.5	73.4	72.5	75.1	76.4	78.7	80.0		
		MCDB	92.0	90.0	90.6	87.0	87.1	83.3	84.5	86.4	84.2	89.0	91.6	91.5		
	2%	WB	79.8	79.8	79.6	77.8	76.1	72.2	71.7	71.0	73.7	75.3	77.4	78.9		
		MCDB	90.6	89.8	89.0	86.2	85.1	82.1	82.4	84.9	85.2	87.9	89.6	90.0		
	5%	WB	79.0	79.0	78.8	77.0	75.0	71.1	70.1	69.6	72.6	74.5	76.5	78.2		
		MCDB	89.4	89.4	87.7	85.2	83.2	81.1	81.2	83.6	85.5	87.5	88.6	88.8		
	10%	WB	78.4	78.3	78.1	76.2	73.9	69.9	68.6	68.3	71.4	73.8	75.8	77.5		
		MCDB	88.1	88.1	86.6	83.9	82.0	80.4	79.7	82.9	85.6	87.2	87.9	87.7		
(35) Mean Daily Temperature Range	MDBR		16.7	17.3	18.2	21.4	23.1	24.7	27.3	30.3	30.2	27.1	23.6	20.1		
	5% DB	MCDBR	20.5	21.1	22.1	24.5	24.4	24.9	28.3	31.3	31.6	28.1	26.3	24.2		
		MCWBR	5.4	5.5	5.8	7.5	7.9	8.4	9.6	10.0	8.8	6.9	6.0	5.5		
	5% WB	MCDBR	16.8	17.1	17.9	18.1	19.1	21.3	23.1	27.0	24.4	24.0	22.0	19.2		
		MCWBR	5.5	5.7	5.5	5.7	6.4	7.3	8.5	8.9	6.5	6.0	5.3	4.8		
(40) Clear-Sky Solar Irradiance	taub		0.404	0.399	0.392	0.374	0.362	0.340	0.335	0.340	0.395	0.417	0.432	0.416		
	taud		2.504	2.518	2.532	2.559	2.556	2.574	2.578	2.535	2.370	2.336	2.346	2.447		
	Ebn at Noon		298	297	293	288	281	283	288	295	288	289	288	294		
	Edh at Noon		36	36	34	32	30	28	29	32	40	43	43	38		
(44) All-Sky Solar Radiation	RadAvg		1865	1868	1816	1779	1640	1567	1671	1923	2142	2237	2154	2020		
	RadStd		165	224	197	185	109	89	68	69	111	127	85	146		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days				
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (4 neighbors)	+0.72	N/A	N/A	N/A	N/A	-0.85	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page